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Homework Set # 5

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$$d^4\sigma = R^2 \frac{d^2W}{\hbar\omega d\Omega d\omega} \sin\theta \cos\theta d\theta d\phi d\Omega d\omega; \text{ Let } \varepsilon = \hbar\omega \text{ s.t. } d\varepsilon = \hbar d\omega$$

$$\Rightarrow \frac{1}{\hbar} \frac{d^4\sigma}{d\Omega d\varepsilon} = \frac{d^2\sigma}{d\Omega d\varepsilon} = \frac{R^2 d^2W}{\hbar^2\omega d\Omega d\omega} \sin\theta \cos\theta d\theta d\phi; \sin\theta \cos\theta = \frac{\sin 2\theta}{2}$$

$$\Rightarrow \frac{d^4\sigma}{d\Omega d\varepsilon} = \frac{R^2 d^2W}{\hbar\varepsilon d\Omega d\omega} \frac{\sin 2\theta}{2} d\theta d\phi$$

$$\text{Then } \frac{d^2\sigma}{d\Omega d\varepsilon} = \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \frac{R^2 d^2W}{\hbar\varepsilon d\Omega d\omega} \frac{\sin 2\theta}{2} = \frac{R^2}{2\hbar\varepsilon} \int_0^{2\pi} d\phi \int_0^{\pi/2} \frac{d^2W}{d\Omega d\omega} \sin 2\theta d\theta$$

$$\frac{d^2W}{d\Omega d\omega} = \frac{D}{16\pi^3} |\hat{n} \times (\vec{\beta}' - \vec{\beta})|^2; \beta' = \beta \text{ (elastic collision)}, \vec{\beta}' - \vec{\beta} = 2(\vec{\beta} \cdot \hat{N})\hat{N}$$

$$|\hat{n} \times (\vec{\beta}' - \vec{\beta})|^2 = |\hat{n} \times 2\hat{N}(\vec{\beta} \cdot \hat{N})|^2 = 4|\hat{n} \times \hat{N} \vec{\beta} \cdot \hat{N}|^2$$

$$= 4[(\hat{n} \cdot \hat{N})(\hat{N} \cdot \hat{N})(\vec{\beta} \cdot \hat{N})^2 - (\hat{n} \cdot \hat{N})^2(\vec{\beta} \cdot \hat{N})^2] = 4\beta^2 \cos^2\theta [1 - (\hat{n} \cdot \hat{N})^2]$$

$$\text{Note: } \hat{N} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

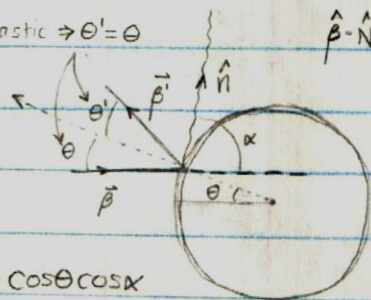
$$\hat{n} = (\sin\chi \cos\phi_\chi, \sin\chi \sin\phi_\chi, \cos\chi)$$

$$\text{Then } \hat{n} \cdot \hat{N} = \sin\theta \cos\phi \sin\chi \cos\phi_\chi + \sin\theta \sin\phi \sin\chi \sin\phi_\chi + \cos\theta \cos\chi$$

$$\frac{d^2W}{d\Omega d\omega} = 4D\beta^2 \cos^2\theta [1 - (\hat{n} \cdot \hat{N})^2]$$

$$\text{Thus, } \frac{d^2\sigma}{d\Omega d\varepsilon} = \frac{4DR^2\beta^2}{2\hbar\varepsilon} \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \sin 2\theta \cos^2\theta [1 - (\sin\theta \cos\phi \sin\chi \cos\phi_\chi + \sin\theta \sin\phi \sin\chi \sin\phi_\chi + \cos\theta \cos\chi)^2]$$

$$= \frac{\mu c q^2 \beta^2 R^2}{48\pi^2 \hbar\varepsilon} (2 + 3\sin^2\chi)$$

elastic $\Rightarrow \theta' = \theta$ $\vec{\beta} \cdot \hat{N} = \cos\theta$ 

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$$10.13 \quad \frac{d^2 W}{d\omega d\Omega} = \frac{D}{4\pi^3 \epsilon_0 c} (\hat{N} \cdot \vec{\beta})^2 |\hat{n} \times \hat{N}|^2 = \frac{D(\hat{N} \cdot \vec{\beta})^2}{4\pi^3 \epsilon_0 c} \sin^2 \theta = D(\hat{N} \cdot \vec{\beta})^2 \sin^2 \theta$$

$$\Rightarrow \frac{dW}{d\omega} = \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta D(\hat{N} \cdot \vec{\beta})^2 \sin^2 \theta = 2\pi D(\hat{N} \cdot \vec{\beta})^2 \int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta$$

$$\int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta = \int_0^{\frac{\pi}{2}} (1 - \cos^2 \theta) \sin \theta d\theta = [\cos \theta]_0^{\frac{\pi}{2}} - \frac{[\cos^3 \theta]_0^{\frac{\pi}{2}}}{3}$$

$$= -[0 - 1] - \frac{[0 - 1]}{3} = \frac{2}{3}$$

$$\therefore \frac{dW}{d\omega} = \frac{2}{3} \times 2\pi \frac{q^2}{4\pi^3 \epsilon_0 c} (\hat{N} \cdot \vec{\beta})^2 = \frac{q^2}{3\pi^2 \epsilon_0 c} (\hat{N} \cdot \vec{\beta})^2 \quad (\text{nonrelativistic})$$

$$\frac{d^2 W}{d\omega d\Omega} = \frac{D_{14}}{16\pi^3 \epsilon_0 c} \frac{|\hat{n} \times \vec{\beta}'|^2}{|1 - \hat{n} \cdot \vec{\beta}'|} = \frac{D}{4} \frac{|\beta' \sin \theta|^2}{|1 - \beta \cos \theta|^2} = \frac{D\beta'^2 \sin^2 \theta}{4(1 - \beta \cos \theta)^2}$$

$$\Rightarrow \frac{dW}{d\omega} = \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{2}} d\theta \frac{D\beta'^2 \sin^2 \theta}{4(1 - \beta \cos \theta)^2} = \frac{-D\pi}{2\beta'} [2\beta' + \beta^2 + 2\ln(1 - \beta')]$$

In the ultrarelativistic limit, $\beta' \approx 1 - \frac{1}{2\gamma^2}$

$$\Rightarrow \frac{dW}{d\omega} \approx \frac{-D\pi}{2 - \frac{1}{\gamma^2}} \left[\left(2 - \frac{1}{\gamma^2}\right) + \left(1 - \frac{1}{2\gamma^2}\right)^2 + 2\ln\left(\frac{1}{2\gamma^2}\right) \right]$$

$$= \frac{-D\pi}{2 - \frac{1}{\gamma^2}} \left[2 - \frac{1}{\gamma^2} + 1 + \frac{1}{4\gamma^4} - \frac{1}{\gamma^2} + 2\ln\left(\frac{1}{2\gamma^2}\right) \right]$$

$$\approx 2 \rightarrow \frac{D\pi}{2} \left[-3 + \frac{2}{\gamma^2} - \frac{1}{4\gamma^4} - 2\ln\left(\frac{1}{2\gamma^2}\right) \right] = D\pi \left[-\frac{3}{2} + \frac{1}{\gamma^2} - \frac{1}{8\gamma^4} - \ln\left(\frac{1}{2\gamma^2}\right) \right]$$

$$= \frac{q^2}{2\pi^2 \epsilon_0 c} \left[\frac{-3}{4} + \frac{2}{4\gamma^2} - \frac{1}{16\gamma^4} - \frac{2}{4} \ln\left(\frac{1}{2\gamma^2}\right) \right] = \frac{q^2}{2\pi^2 \epsilon_0 c} \left[\frac{-3}{4} + \frac{2}{4\gamma^2} - \frac{1}{16\gamma^4} + \ln(\sqrt{2}\gamma) \right]$$

$$= \frac{q^2}{2\pi^2 \epsilon_0 c} \left[\ln \sqrt{2} - \frac{3}{4} + \frac{2}{4\gamma^2} - \frac{1}{16\gamma^4} + \ln \gamma \right] \approx \frac{q^2}{2\pi^2 \epsilon_0 c} \left[\ln \sqrt{2} - \frac{3}{4} + \ln \gamma \right]$$

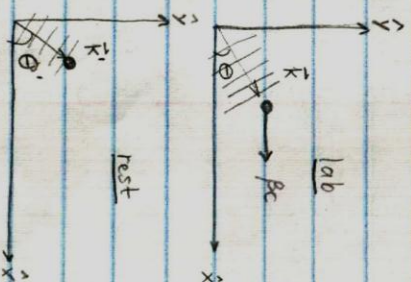
$$\ln \sqrt{2} - \frac{3}{4} \approx .40 \ll \ln \gamma \Rightarrow \frac{dW}{d\omega} \approx \frac{q^2}{2\pi^2 \epsilon_0 c} \ln \gamma$$

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a) $E_{\text{photon}} = pc \Rightarrow p_{\text{rester}} = \frac{dE_{\text{photon}}}{dt} = \frac{cdp}{dt}$ (in the rest frame)

In the rest frame, $P = \vec{v}_T S_{\text{rest}} \Rightarrow \frac{c(dP)}{(dt)_{\text{rest}}} = \vec{v}_T S_{\text{rest}}$

Transforming to the rest frame:

$$\begin{bmatrix} \omega'/c \\ k'_x \\ k'_y \\ k'_z \end{bmatrix} = \begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \omega/c \\ k_x \\ k_y \\ k_z \end{bmatrix} = \begin{bmatrix} \gamma \frac{\omega}{c} - \beta\gamma k_x \\ -\beta\gamma \frac{\omega}{c} + \gamma k_x \\ k_y \\ k_z \end{bmatrix} = \gamma \begin{bmatrix} \frac{\omega}{c} - \beta k_x \\ k_x - \beta \frac{\omega}{c} \\ \frac{1}{\gamma} k_y \\ \frac{1}{\gamma} k_z \end{bmatrix}$$



$$\frac{\omega'}{c} = \gamma \left[\frac{\omega}{c} - \beta k \cos\theta \right] = \gamma \frac{\omega}{c} [1 - \beta \cos\theta] \Rightarrow \omega' = \gamma \omega (1 - \beta \cos\theta)$$

$$k'_x = \gamma \left[k \cos\theta - \beta \frac{\omega}{c} \right] = \gamma \frac{\omega}{c} [\cos\theta - \beta] \Rightarrow k'_x = \gamma k (\cos\theta - \beta)$$

$$k'_y = k \sin\theta$$

$$S \equiv \frac{\omega^2 |A|^2}{4\pi c} \Rightarrow S' = \frac{\omega'^2 |A|^2}{4\pi c} = \frac{\gamma^2 \omega^2 (1 - \beta \cos\theta)^2 |A|^2}{4\pi c} = \gamma^2 (1 - \beta \cos\theta)^2 S$$

Then, from above, $\frac{c(dP)}{(dt)_{\text{rest}}} = \vec{v}_T S' = \vec{v}_T \gamma^2 (1 - \beta \cos\theta)^2 S$

$$\Rightarrow \frac{dP}{dt} = \frac{\gamma^2 \vec{v}_T S}{c} (1 - \beta \cos\theta)^2$$

$$\left(\frac{dP}{dt} \right)_x = \frac{dP}{dt} \cos\theta' ; \left(\frac{dP}{dt} \right)_y = \frac{dP}{dt} \sin\theta'$$

$$\sin\theta' = \frac{k'_y}{k'} = \frac{k \sin\theta}{\gamma \sqrt{\gamma^2 (\cos\theta - \beta)^2 \sin^2\theta + \sin^2\theta}} = \frac{\sin\theta}{\gamma \sqrt{\cos^2\theta + \gamma^2 \beta^2 - 2\beta\gamma^2 \cos\theta + \sin^2\theta}}$$

$$\begin{aligned} &= \gamma \left[\cos^2\theta + \gamma^2 \beta^2 - 2\beta\gamma^2 \cos\theta + \sin^2\theta \right]^{-1/2} = \frac{1}{\sqrt{1 - \beta^2}} \left[\cos^2\theta + \beta^2 - 2\beta\gamma^2 \cos\theta + (1 - \beta^2)(1 - \cos^2\theta) \right]^{-1/2} \\ &= \gamma \left[1 + \beta^2 - \beta^2 + \cos^2\theta - \cos^2\theta - 2\beta\gamma^2 \cos\theta + \beta^2 \cos^2\theta \right]^{-1/2} = \gamma \left[1 - 2\beta\gamma^2 \cos\theta + \beta^2 \cos^2\theta \right]^{-1/2} \\ &= \gamma \left[(1 - \beta \cos\theta)^2 \right]^{-1/2} = \gamma (1 - \beta \cos\theta) \Rightarrow \sin\theta' = \frac{\sin\theta}{\gamma (1 - \beta \cos\theta)} \end{aligned}$$

$$\cos \theta' = \frac{k_x}{k} = \frac{\gamma k (\cos \theta - \beta)}{k(1 - \beta \cos \theta)} = \frac{\cos \theta - \beta}{(1 - \beta \cos \theta)}$$

$$\therefore \left(\frac{dp}{dt} \right)_x = \frac{\gamma^2 S}{c} (1 - \beta \cos \theta)^2 (\cos \theta - \beta) = \frac{\gamma^2 S}{c} (1 - \beta \cos \theta) (\cos \theta - \beta)$$

$$\left(\frac{dp}{dt} \right)_y = \frac{\gamma^2 S}{c} (1 - \beta \cos \theta)^2 \sin \theta = \frac{\gamma^2 S}{c} (1 - \beta \cos \theta) \sin \theta$$

$$b) \frac{dp}{dt} = \frac{\gamma^2 S}{c} = \frac{\gamma^2 S}{c} \rightarrow \int dp = \frac{\gamma^2}{c} \int S dt \rightarrow P = \frac{\gamma^2}{c} \int S$$

$$v \approx \frac{v_T}{mc} \int = \frac{6.65 \times 10^{-29}}{9.11 \times 10^{-31} \times 10^8} \quad 10^6 = \frac{6.65}{27.33} \frac{10^{-19}}{10^{-23}} = 2433 \frac{m}{s}$$

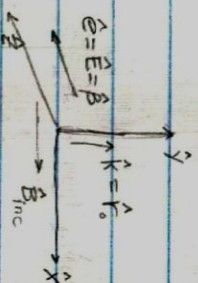
$$\vec{B}(\vec{r}_0, t_0) \stackrel{\text{for field}}{=} \frac{-\mu_0 q}{4\pi} \left\{ \frac{\hat{r}_0 \times [\vec{\beta} + \hat{r}_0 \times (\vec{\beta} \times \vec{\beta})]}{R_0 (1 - \hat{r}_0 \cdot \vec{\beta})^3} \right\} \approx \frac{-\mu_0 q}{4\pi} \left\{ \frac{\hat{r}_0 \times [\vec{\beta} + \hat{r}_0 \times (\vec{\beta} \times \vec{\beta})]}{r (1 - \hat{r}_0 \cdot \vec{\beta})^3} \right\}$$

For the incident wave, let $\hat{e} = \hat{z}$, $\hat{k} = \hat{y}$

$$\vec{B}_{inc} = \vec{\nabla} \times \vec{A} = A_0 \hat{x} \left(\frac{\partial A_z}{\partial y} \right) = A_0 \hat{x} \frac{\partial}{\partial y} e^{i(ky - \omega t)} = A_0 \hat{x} \frac{\partial}{\partial y} e^{i(ky - \omega t)} = i k A_0 e^{i(ky - \omega t)} \hat{x}$$

For the particle experiencing an $\vec{E}$ -field:

$$\vec{F} = m\vec{a} \rightarrow \vec{E} = q = m c \dot{\vec{\beta}} \Rightarrow \dot{\vec{\beta}} = -\frac{q}{mc} \frac{\partial \vec{A}}{\partial t} \quad \hat{e} = \vec{E} = \vec{\beta} \quad \hat{k} = \hat{y}$$



$$\text{Then } \dot{\vec{\beta}} = -\frac{q}{mc} \frac{\partial \hat{z} A_0 e^{i(ky - \omega t)}}{\partial t} = +\frac{i\omega q A_0}{mc} e^{i(ky - \omega t)} \hat{z} \quad \text{we also know that}$$

$$\vec{\beta} \parallel \dot{\vec{\beta}} \Rightarrow \vec{\beta} \times \dot{\vec{\beta}} = 0.$$

$$\text{In the Liénard-Wiechert Thm above, we see } \vec{B}_{rad} = -\frac{\mu_0 q}{4\pi} \frac{\vec{r}_0 \times \dot{\vec{\beta}}}{r_0 (1 - \hat{r}_0 \cdot \vec{\beta})^3}$$

Since we're looking behind the particle, $\hat{r}_0 = \hat{k} = \hat{y} \Rightarrow \hat{r}_0 \times \dot{\vec{\beta}} = \dot{\vec{\beta}} \times \hat{x}$

$$\hat{r}_0 \cdot \dot{\vec{\beta}} = 0 \Rightarrow \vec{B}_{rad} = -\frac{\mu_0 q}{4\pi} \frac{\dot{\vec{\beta}} \times \hat{x}}{r_0} = -\frac{\mu_0 q}{4\pi} \frac{\dot{\vec{\beta}} \times \hat{x}}{r_0} = -\frac{\mu_0 q}{4\pi} \frac{i\omega q A_0 e^{i(ky - \omega t)}}{mc y}$$

$$\therefore |\vec{B}_{total}| = |\vec{B}_{inc} + \vec{B}_{rad}| = i k A_0 e^{i(ky - \omega t)} \left[1 - \frac{\mu_0 q^2}{4\pi m y} \right] < |\vec{B}_{inc}| \Rightarrow \text{destructive interference}$$